

ECE 726 - Assignment 1

Trade-off between Overfitting and Underfitting

Mahan Choudhury
Student ID: 400648097

January 28, 2026

Contents

1	Introduction	2
1.1	Problem Setup	2
1.2	Data Generation	2
2	Polynomial Regression Without Regularization	2
2.1	Methodology	2
2.2	RMSE Results	2
2.3	RMSE vs Polynomial Degree Plot	3
2.4	Analysis of Overfitting and Underfitting Regions	3
2.4.1	Underfitting Region ($M = 1, 2$)	3
2.4.2	Optimal Capacity Region ($M = 3-6$)	4
2.4.3	Overfitting Region ($M = 7, 8$)	4
2.5	Polynomial Fit Visualizations	4
3	L_2 Regularization ($M = 8$)	5
3.1	Methodology	5
3.2	RMSE vs Regularization Strength	6
3.3	Analysis of Regularization Regions	6
3.3.1	Overfitting Region ($\log_{10}(\lambda) < -8$)	6
3.3.2	Optimal Capacity Region ($\log_{10}(\lambda) \approx -7$ to -4)	6
3.3.3	Underfitting Region ($\log_{10}(\lambda) > -2$)	6
3.4	Selected Regularization Values	7
3.5	Weight Vector Analysis	7
3.5.1	$\lambda_1 = 10^{-14}$ (Overfitting Region)	7
3.5.2	$\lambda_2 = 10^{-6}$ (Optimal Region)	7
3.5.3	$\lambda_3 = 10^2$ (Underfitting Region)	8
3.5.4	Summary of Weight Behavior	8
3.6	Regularized Fit Visualizations	9
4	Best Model Selection and Test Performance	9
4.1	Model Comparison	9
4.2	Selected Model	9
4.3	Test Set Performance	10
5	Conclusions	10

1 Introduction

The bias-variance trade-off is a fundamental concept in machine learning that describes the relationship between model complexity and generalization error. Models that are too simple suffer from **underfitting** (high bias), while models that are too complex suffer from **overfitting** (high variance). The goal is to find the optimal model complexity that minimizes the total error on unseen data.

1.1 Problem Setup

The true underlying function is:

$$f_{\text{true}}(x) = \sin(2\pi x) \quad (1)$$

Observations are generated according to:

$$t = \sin(2\pi x) + \epsilon, \quad \text{where } \epsilon \sim \mathcal{N}(0, 0.04) \quad (2)$$

The noise has standard deviation $\sigma = \sqrt{0.04} = 0.2$. The optimal predictor that minimizes expected squared error is:

$$f_{\text{opt}}(x) = \mathbb{E}[t|x] = \sin(2\pi x) \quad (3)$$

1.2 Data Generation

Using random seed **8097** (last 4 digits of student ID), we generated:

- **Training set:** 9 samples with x uniformly spaced in $[0, 1]$
- **Validation set:** 100 samples with x uniformly spaced in $[0, 1]$
- **Test set:** 100 samples with x uniformly spaced in $[0, 1]$

2 Polynomial Regression Without Regularization

2.1 Methodology

We trained polynomial models of the form:

$$f_M(x) = w_0 + w_1x + w_2x^2 + \dots + w_Mx^M = \sum_{j=0}^M w_jx^j \quad (4)$$

for $M = 1, 2, \dots, 8$ (since $N-1 = 8$). This can be viewed as linear regression with polynomial features $\phi(x) = [1, x, x^2, \dots, x^M]^T$.

The design matrix Φ has rows $\phi(x^{(i)})^T$, and the optimal weights are computed using the closed-form solution (normal equation):

$$\mathbf{w}^* = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{t} \quad (5)$$

2.2 RMSE Results

Table 1 shows the training and validation Root Mean Squared Error (RMSE) for each polynomial degree.

Table 1: Training and Validation RMSE vs Polynomial Degree M

M	Training RMSE	Validation RMSE
1	0.4752	0.5262
2	0.4566	0.5477
3	0.1796	0.2721
4	0.1183	0.2799
5	0.1056	0.2719
6	0.0934	0.2672
7	0.0658	0.2923
8	0.0000	0.4161

Optimal Predictor RMSE (Validation): 0.1838

This represents the irreducible error due to noise. Since variance = 0.04, the expected RMSE $\approx \sqrt{0.04} = 0.2$, which matches our observation.

2.3 RMSE vs Polynomial Degree Plot

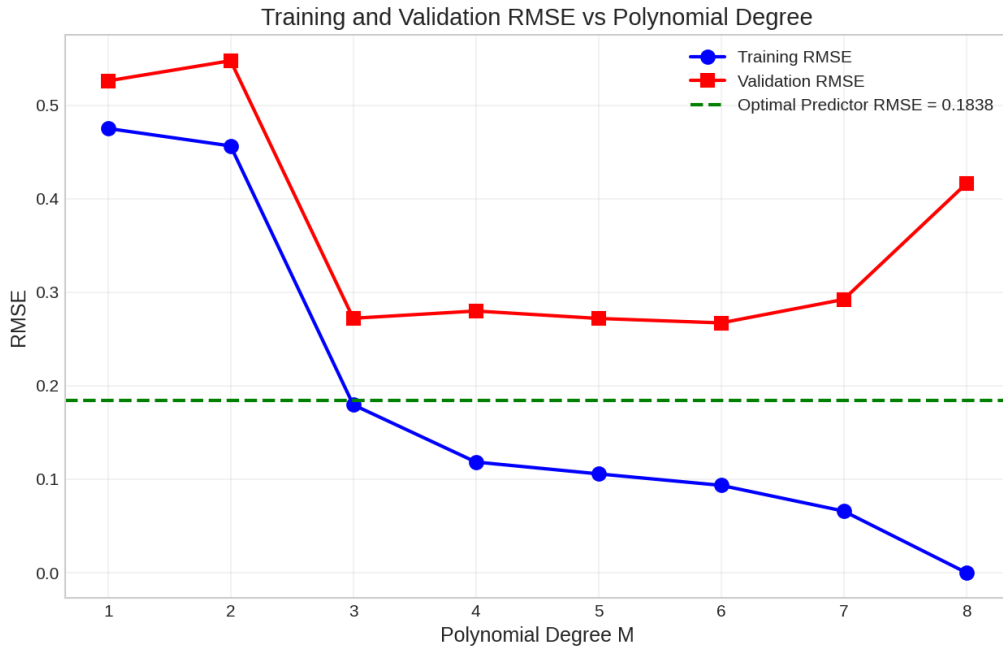


Figure 1: Training and Validation RMSE versus polynomial degree M . The green dashed line shows the optimal predictor's RMSE, representing the irreducible error.

2.4 Analysis of Overfitting and Underfitting Regions

Based on Figure 1 and Table 1, we identify three regions:

2.4.1 Underfitting Region ($M = 1, 2$)

Both training and validation RMSE are high (≈ 0.45 – 0.55). The linear and quadratic models are too simple to capture the sinusoidal pattern. The model has **high bias**, thus it cannot represent the underlying function regardless of the training data.

2.4.2 Optimal Capacity Region ($M = 3\text{--}6$)

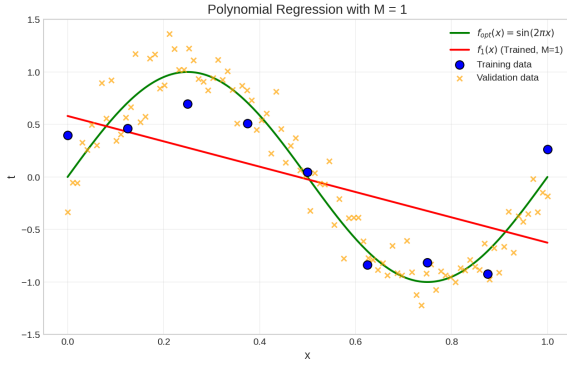
The validation RMSE reaches its minimum at $M = 6$ (RMSE = 0.2672). The models capture the essential sinusoidal shape while not fitting noise excessively. This represents the best **bias-variance trade-off**.

2.4.3 Overfitting Region ($M = 7, 8$)

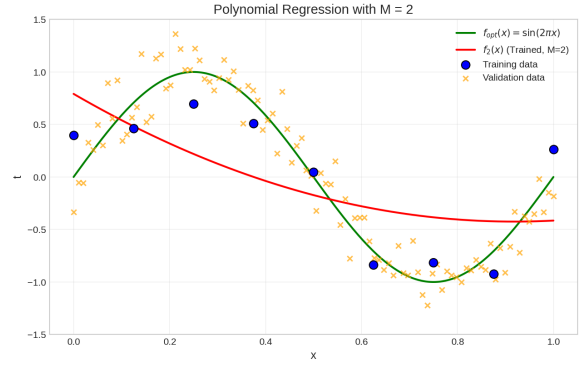
Training RMSE drops dramatically (0.066 for $M = 7$, 0.000 for $M = 8$), but validation RMSE increases. At $M = 8$ with 9 training points, the polynomial passes through all training points exactly (perfect interpolation), but the model has learned the noise rather than the underlying pattern. This is **high variance**; for this reason the model is too sensitive to specific training examples.

2.5 Polynomial Fit Visualizations

The following figures show the fitted polynomials (red) compared to the true function (green) for all values of M from 1 to 8. Training data points are shown as blue circles and validation data as orange crosses.

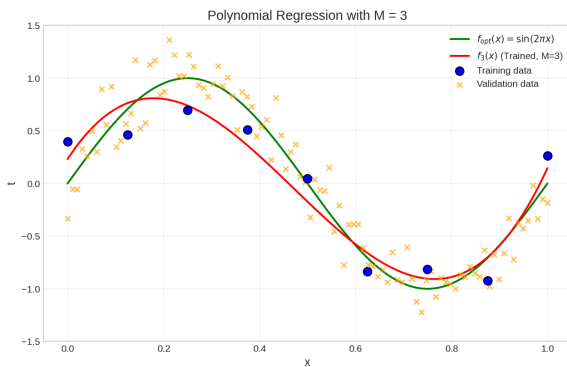


(a) $M = 1$ (Underfitting)

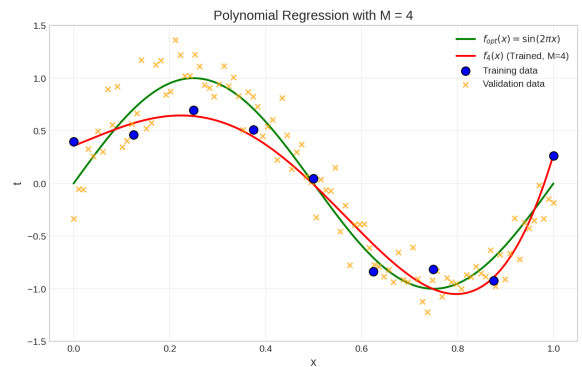


(b) $M = 2$ (Underfitting)

Figure 2: Polynomial fits for $M = 1$ and $M = 2$. Both models are too simple to capture the sinusoidal pattern — this is the **underfitting** region.

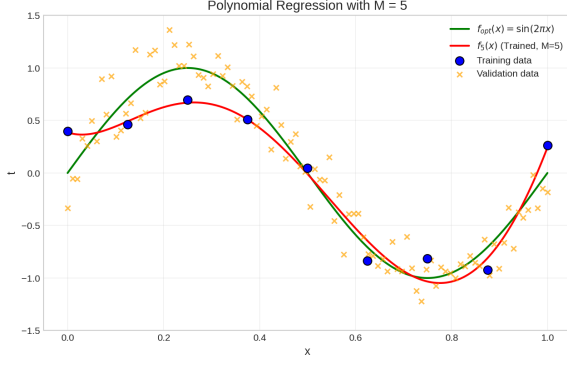


(a) $M = 3$

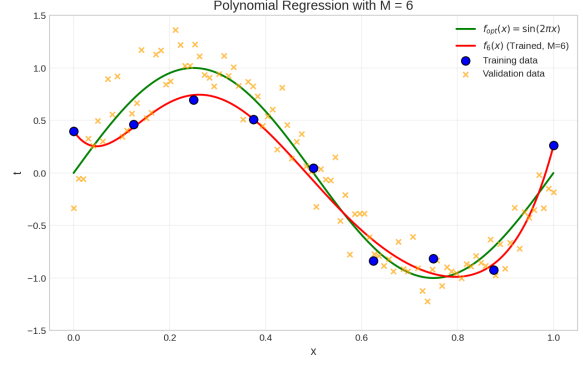


(b) $M = 4$

Figure 3: Polynomial fits for $M = 3$ and $M = 4$. The models begin to capture the sinusoidal shape reasonably well.

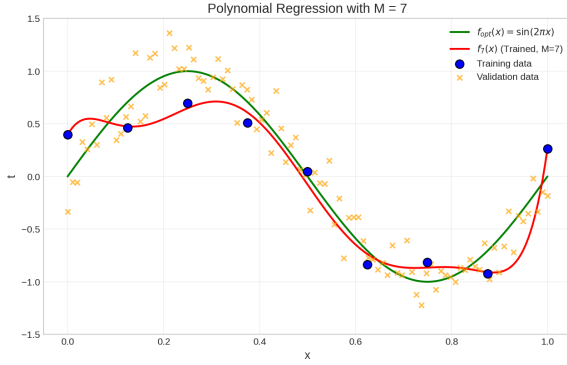


(a) $M = 5$

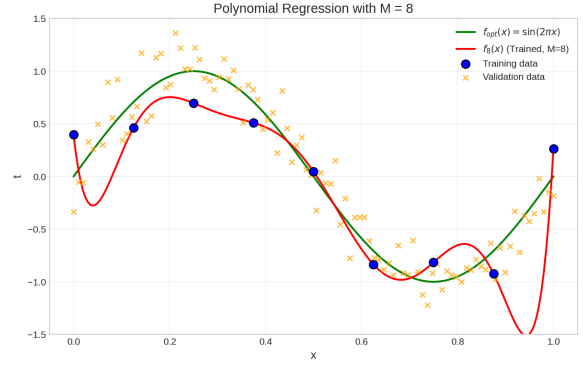


(b) $M = 6$ (Best without regularization)

Figure 4: Polynomial fits for $M = 5$ and $M = 6$. These are in the **optimal capacity region**. $M = 6$ achieves the lowest validation RMSE among unregularized models.



(a) $M = 7$ (Beginning to overfit)



(b) $M = 8$ (Severe overfitting)

Figure 5: Polynomial fits for $M = 7$ and $M = 8$. This is the **overfitting region**. At $M = 8$, the polynomial passes through all 9 training points exactly (training RMSE = 0), but shows wild oscillations between points, especially near the boundaries.

Observations from the visualizations:

- $M = 1, 2$: The curves are too simple (linear/quadratic) to approximate a sine wave.
- $M = 3-6$: The polynomials follow the sinusoidal pattern increasingly well.
- $M = 7, 8$: The curves begin to oscillate wildly, fitting the noise rather than the underlying pattern. At $M = 8$, the polynomial interpolates all training points perfectly but generalizes poorly.

3 L_2 Regularization ($M = 8$)

3.1 Methodology

To mitigate overfitting with the highest-capacity model ($M = 8$), we applied L_2 regularization (Ridge Regression). The regularized loss function is:

$$L(\mathbf{w}) = \frac{1}{N} \sum_{i=1}^N \left(t^{(i)} - f(\mathbf{x}^{(i)}) \right)^2 + \lambda \|\mathbf{w}\|^2 \quad (6)$$

The closed-form solution becomes:

$$\mathbf{w}^* = (\Phi^T \Phi + \lambda \mathbf{I})^{-1} \Phi^T \mathbf{t} \quad (7)$$

Features were standardized (zero mean, unit variance) before training with regularization, as required for proper regularization behavior.

We tested $\lambda = 10^i$ for all integers i from -14 to 2 inclusive.

3.2 RMSE vs Regularization Strength

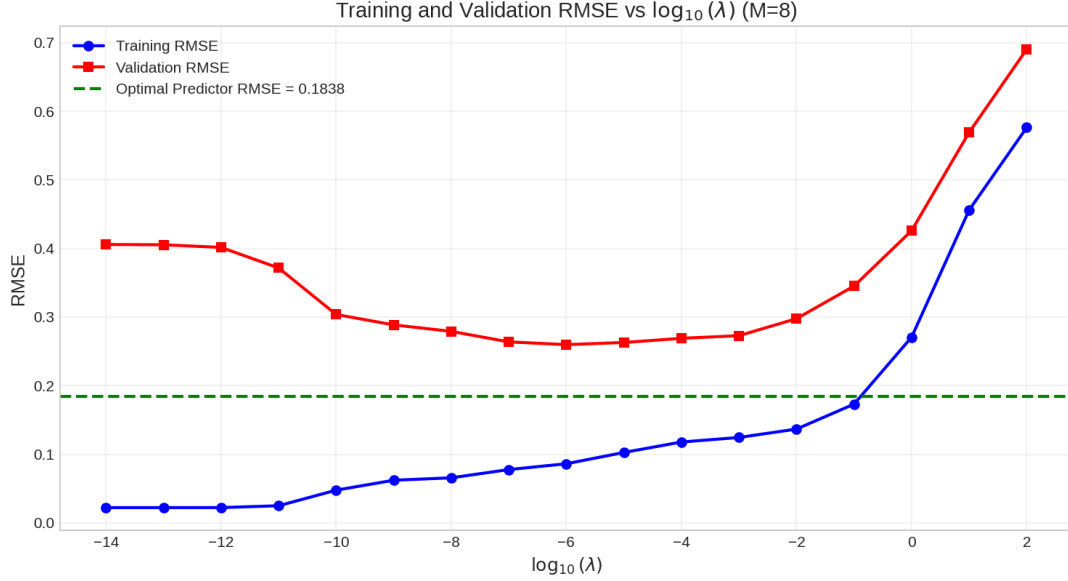


Figure 6: Training and Validation RMSE versus $\log_{10}(\lambda)$ for $M = 8$. The green dashed line shows the optimal predictor's RMSE.

3.3 Analysis of Regularization Regions

3.3.1 Overfitting Region ($\log_{10}(\lambda) < -8$)

When λ is very small (weak regularization), the model behaves similarly to the unregularized case. Training RMSE is very low (≈ 0.02), but validation RMSE is high (≈ 0.40). The regularization term has minimal effect on the weights.

3.3.2 Optimal Capacity Region ($\log_{10}(\lambda) \approx -7$ to -4)

The validation RMSE reaches its minimum at $\log_{10}(\lambda) = -6$ (i.e., $\lambda = 10^{-6}$) with RMSE = 0.2597. This is the optimal regularization strength that balances fitting the data and controlling model complexity.

3.3.3 Underfitting Region ($\log_{10}(\lambda) > -2$)

When λ is large (strong regularization), the weights are heavily penalized and shrink toward zero. Both training and validation RMSE increase significantly. At $\lambda = 100$, the model essentially becomes a nearly constant function.

3.4 Selected Regularization Values

Table 2 shows the three selected λ values representing each region.

Table 2: Selected λ Values from Different Regions

Region	λ	$\log_{10}(\lambda)$	Validation RMSE	$\ \mathbf{w}\ $
Overfitting	10^{-14}	-14	0.4057	27504.17
Optimal	10^{-6}	-6	0.2597	98.51
Underfitting	10^2	2	0.6901	0.042

3.5 Weight Vector Analysis

The weight vector norms clearly demonstrate the effect of regularization. Below are the complete weight vectors $\mathbf{w} = [w_0, w_1, w_2, w_3, w_4, w_5, w_6, w_7, w_8]$ for each selected λ value.

3.5.1 $\lambda_1 = 10^{-14}$ (Overfitting Region)

$$\mathbf{w}_1 = \begin{bmatrix} 0.0000 \\ -13.0323 \\ 265.2122 \\ -1940.4891 \\ 7245.4458 \\ -15236.0326 \\ 18151.0923 \\ -11400.5903 \\ 2928.2479 \end{bmatrix}, \quad \|\mathbf{w}_1\| = 27504.17$$

The weights are **extremely large**, with magnitudes in the thousands. This allows the polynomial to make sharp oscillations to pass through each training point exactly, resulting in overfitting.

3.5.2 $\lambda_2 = 10^{-6}$ (Optimal Region)

$$\mathbf{w}_2 = \begin{bmatrix} 0.0000 \\ -1.2095 \\ 16.8180 \\ -47.5351 \\ 20.2371 \\ 46.0728 \\ -12.2658 \\ -57.1251 \\ 34.8534 \end{bmatrix}, \quad \|\mathbf{w}_2\| = 98.51$$

The weights are **moderate**, with magnitudes mostly between 1 and 60. This provides a good balance between fitting the training data and maintaining smoothness, resulting in the best generalization.

3.5.3 $\lambda_3 = 10^2$ (Underfitting Region)

$$\mathbf{w}_3 = \begin{bmatrix} 0.0000 \\ -0.0278 \\ -0.0230 \\ -0.0167 \\ -0.0111 \\ -0.0065 \\ -0.0027 \\ 0.0002 \\ 0.0026 \end{bmatrix}, \quad \|\mathbf{w}_3\| = 0.042$$

The weights are **nearly zero**, all with magnitudes less than 0.03. The strong regularization has shrunk the weights so much that the model becomes essentially a flat line, unable to capture any pattern.

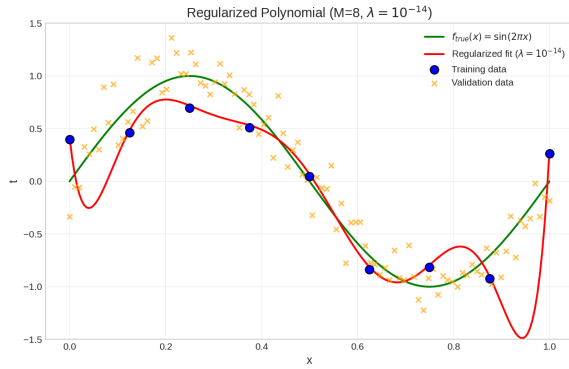
3.5.4 Summary of Weight Behavior

Table 3: Comparison of Weight Vector Properties

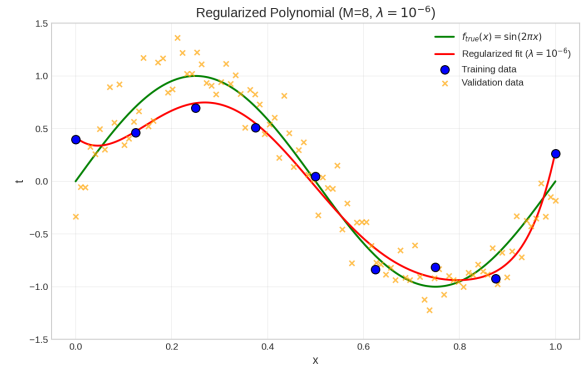
Property	$\lambda_1 = 10^{-14}$	$\lambda_2 = 10^{-6}$	$\lambda_3 = 10^2$
$\ \mathbf{w}\ $	27504.17	98.51	0.042
Max $ w_i $	18151.09	57.13	0.028
Behavior	Wild oscillations	Smooth curve	Nearly flat
Region	Overfitting	Optimal	Underfitting

These results confirm our theoretical understanding: regularization controls model complexity by penalizing large weights. The optimal λ finds the right balance where weights are large enough to fit the underlying pattern but small enough to avoid fitting noise.

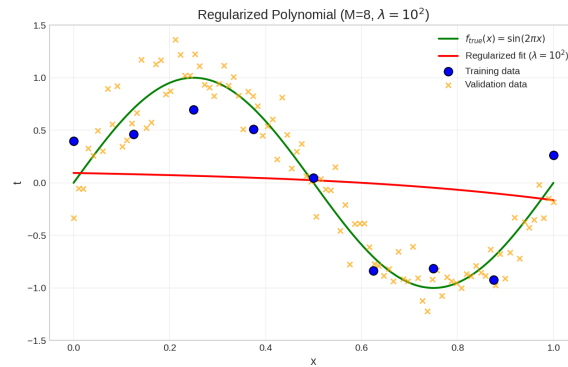
3.6 Regularized Fit Visualizations



(a) $\lambda = 10^{-14}$ (Overfitting)



(b) $\lambda = 10^{-6}$ (Optimal)



(c) $\lambda = 10^2$ (Underfitting)

Figure 7: Regularized polynomial regression fits ($M = 8$) for three values of λ . Green curve: true function $f_{\text{true}}(x) = \sin(2\pi x)$. Red curve: regularized fit.

4 Best Model Selection and Test Performance

4.1 Model Comparison

Comparing the best models from each approach:

- **Best polynomial model** ($M = 6$, no regularization): Validation RMSE = 0.2672
- **Best regularized model** ($M = 8, \lambda = 10^{-6}$): Validation RMSE = 0.2597

4.2 Selected Model

The **regularized polynomial with $M = 8$ and $\lambda = 10^{-6}$** is selected as the best model because it achieves the lowest validation RMSE among all models trained.

4.3 Test Set Performance

Table 4: Final Test Set Performance

Metric	Value
Best Model Test RMSE	0.2472
Optimal Predictor Test RMSE	0.1802
Difference from Optimal	0.0670

The selected model achieves a test RMSE of **0.2472**, which is reasonably close to the irreducible error of approximately 0.18–0.20.

5 Conclusions

This experiment successfully demonstrated the bias-variance trade-off in polynomial regression:

1. **Underfitting vs. Overfitting:** Models that are too simple (low M or high λ) suffer from high bias and cannot capture the underlying pattern. Models that are too complex (high M or low λ) suffer from high variance and fit noise in the training data.
2. **Optimal Capacity:** The best generalization is achieved at intermediate complexity levels. For polynomial degree, $M = 6$ performed best among unregularized models. With regularization, $\lambda = 10^{-6}$ achieved even better validation performance.
3. **Regularization Effectiveness:** L_2 regularization allows using a high-capacity model ($M = 8$) while preventing overfitting by penalizing large weights. This provides a more flexible approach to capacity control than simply limiting polynomial degree.
4. **Irreducible Error:** The optimal predictor’s RMSE (≈ 0.18 – 0.20) represents the irreducible error due to noise in the data. No model can achieve lower error on average.
5. **Consistency of Observations:** The error curves (Figures 1 and 6) are fully consistent with the prediction function plots (Figures 2-5 and 7). Models in the overfitting region show wild oscillations, while models in the underfitting region cannot follow the sinusoidal pattern.
6. **Weight Vector Behavior:** The weight vector norms confirm our understanding: overfitting corresponds to large weights ($\|\mathbf{w}\| \approx 27504$), optimal capacity to moderate weights ($\|\mathbf{w}\| \approx 99$), and underfitting to very small weights ($\|\mathbf{w}\| \approx 0.04$).